Belief Revision for Reasoning About Machine Learning

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Abstract

1 Technical Preliminaries

We consider a multi-class classification setting with domain incremental learning style setup. Let $\mathcal{X}$ denote the space of all vectors (raw data) and we will often use x to denote a single vector, X to denote a set ("batch") of samples, and D to denote a distribution over $\mathcal{X}$. We assume access to ground truth oracle $g : \mathcal{X} \to \mathcal{S}$. This oracle is used to train/re-train a machine learning model $F_\Theta : \mathcal{X} \to \mathcal{C}$, where $\mathcal{C} = \{c_1, \ldots, c_n\}$ is the set of of $n > 1$ possible classes or classification labels, and we refer to Θ as a "configuration" for F_Θ[1].

To formalize the incremental learning setup, we adopt some ideas from the classic work on prompt-based continual learning [1, 2] and use the notation $\mathcal{P}$ to denote a set (pool) of discrete objects or prompts, to which input data is mapped to. Note that we do not distinguish between the prompt and prompt key in a key-query pair as in [1]. In that in that paper, and the follow-on work, a query function is applied to the input and returns a key that is associated with a given prompt from the prompt pool $\mathcal{P}$. This is a notational convenience that we employ due to the fact that in this work, we simply treat prompts as discrete objects and do not modify the way they are used with respect to a machine learning model, but we still can capture the full framework of prompt-based learning like [1]. In this way, without loss of generalization, we can assume $\mathcal{P}$ is a finite set of propositions, and will denote with $\mathsf{formula}(\mathcal{P})$ the set of w.f.f. that can be created through the classical connectives $\neg, \wedge, \vee$ over the propositions in $\mathcal{P}$.

Note that other works in continual learning [3–6] have used this same idea as well as those that we formalize as a configuration Θ in Definition 1.1.

[1] Given an example of the type data that the configuration encodes?

We also assume two more finite (?) sets of propositions, $\mathcal{F}$ and $\mathcal{Q}$, where each $f_i \in \mathcal{F}$ is a symbol that represents a learning model $f_i : \mathcal{X} \to \mathcal{C}$ and each $q_j \in \mathcal{Q}$ is a symbol that represents a query function $q_j : \mathcal{X} \to 2^{\mathcal{P}}$. Note that we assume $\mathcal{F}, \mathcal{Q}, \mathcal{P}$ are all disjoint.

Example 1.1. *For our running/toy example, let* $\mathcal{F} = \{f_1, f_2, f_3\}$, $\mathcal{C} = \{c_1, c_2\}$, $\mathcal{Q} = \{q_1, q_2, q_3, q_4\}$, *and* $\mathcal{P} = \{p_1, p_2, p_3, p_4, p_5\}$.

	Learning Models f_i			Query Functions q_j			
x	$f_1(x)$	$f_2(x)$	$f_3(x)$	$q_1(x)$	$q_2(x)$	$q_3(x)$	$q_4(x)$
x_1	c_1	c_1	c_2	p_1	p_1	p_4	p_5
x_2	c_2	c_1	c_1	p_2	p_3	p_5	p_5

Table 1: Values of functions f_i and q_j for vectors $x_1, x_2 \in \mathcal{X}$.

We define a *selection rule* as an expression of the following form $f \leftarrow w$, where $f \in \mathcal{F}$ and $w \in \mathsf{formula}(\mathcal{P})$.

Example 1.2. *For instance, selection rules for our running example could be* $r_1 : f_1 \leftarrow p_2 \wedge p_4$, $r_2 : f_2 \leftarrow p_3 \vee p_5$, *and* $r_3 : f_3 \leftarrow p_1$.

Given a formula $w \in \mathsf{frml}$ and a set of prompts $\mathcal{P}'$, we denote with $\mathcal{P}' \models w$ whether $\mathcal{P}'$ satisfies w is satisfied, in the classical sense.

Finally, given a set of selection rules Π and a set of prompts $\mathcal{P}' \subseteq \mathcal{P}$, let $\Gamma_\Pi(\mathcal{P}')$ be the set of all model symbols from $f_i \in \mathcal{F}$ such that there exists a rule in $f_i \leftarrow w \in \Pi$ such that $\mathcal{P}' \models w$. We will abuse notation and use Γ such that $\Gamma_\Pi(\mathcal{P}')(x) = \{c \in \mathcal{C} \mid f(x) = c \text{ and } f \in \Gamma_\Pi(\mathcal{P}')\}$.

Example 1.3. *Given the set* $\mathcal{P}' = \{p_1, p_2, p_4\}$ *and* $\Pi = \{r_1, r_2, r_3\}$, *we have that* $\Gamma_\Pi(\mathcal{P}') = \{f_1, f_3\}$. *Therefore,* $\Gamma_\Pi(\mathcal{P}')(x_1) = \{f_1(x_1), f_3(x_1)\} = \{c_1, c_2\}$.

Definition 1.1 (Configuration)**.** *Configuration* $\Theta = \langle \Psi, \Pi \rangle$ *is a tuple consisting of set of model-query function pairs,* $\Psi = \{\langle f_1, q_1 \rangle, \ldots, \langle f_n, q_n \rangle\}$ *and a set of selection rules* Π. *We will use* $\mathcal{F}_\Theta(x)$ *to denote the set* $\bigcup_i q_i(x)$. *Given some vector* x, $F_\Theta(x) = \Gamma_\Pi(\mathcal{F}_\Theta(x))(x)$.

Note that for a given $x \in \mathcal{X}$ and Π, we assume $\Gamma_\Pi(\mathcal{F}_\Theta(x))$ is expected to return a single function (otherwise Π is considered inconsistent).

MVM: To MVM: Provide a notion of consistency.

MVM: To MVM: Better example, for instance q could return a set of prompts.

Example 1.4. *Consider our running example with* $\Pi = \{r_1, r_2, r_3\}$, *let* $\Theta = \langle \{\langle f_1, q_1 \rangle, \langle f_2, q_2 \rangle, \langle f_3, q_3 \rangle\} \rangle$. *We have* $\mathcal{F}_\Theta(x_1) = \bigcup_{1 \le i \le 3} q_i(x_1) = \{p_1, p_4\}$, $\mathcal{F}_\Theta(x_2) = \bigcup_{1 \le i \le 3} q_i(x_2) = \{p_2, p_3, p_5\}$.

$\Gamma_\Pi(\mathcal{F}_\Theta(x_1)) = \Gamma_\Pi(\{p_1, p_4\} = \{f_3\}$, *and therefore* $F_\Theta(x_1) = c_2$.

$\Gamma_\Pi(\mathcal{F}_\Theta(x_2)) = \Gamma_\Pi(\{p_2, p_3, p_5\} = f_2$, *and therefore* $F_\Theta(x_2) = c_1$.

Query functions are commonly used in continual learning [1, 2], though we note that the learning models in those works accept the result of the query function (which is also a set of prompts) and the original input vector as arguments. However, we can easily encode this in our framework as follows, f_i in a $\langle f_i, q_i \rangle$ is defined as $f_i(x) = f'(x, q_i(x))$ where f' is a model specified in one of these paradigms (e.g., L2P [1]).

We also define the function tng which takes a batch $X \subseteq \mathcal{X}$ and returns a model-query function pair, e.g., $\mathsf{tng}(X) = \langle f, q \rangle$. We can also represent a configuration for a newly-trained model as $\langle \{\langle f, q \rangle\}, \{f \leftarrow\} \rangle$, where $f \leftarrow$ denotes a rule with a tautology as a body.

Throughout this paper, we will assume that for any $\Psi = \{\langle f_1, q_1 \rangle, \ldots, \langle f_n, q_n \rangle\}$, that the associated query functions will all have disjoint ranges. In other words, for each q_i there is a separate $\mathcal{P}_i \subset \mathcal{P}$ associated with it, and no $p \in \mathcal{P}_i$ can be returned in the output of any other query function. Practically, this is trivial to enforce, as we always know which query function returned a given prompt (as query functions in the continual learning literature are themselves trained models).

We will also assume that for any Π in a configuration prior to revision is consistent. Here we have an implicit constraint that there is no pair of rules $f \leftarrow \mathit{body}, f' \leftarrow \mathit{body}$ where $f \neq f'$ and $\mathcal{F}_\Theta(x) \models \mathit{body} \wedge \mathit{body}'$.

In practice, this is avoidable as query functions in continual learning return a fixed-size set of prompts (and we can add to that set negations for the non-selected prompts), which enable the creation of rule templates to ensure consistency. Note that we will not assume consistency post-revision as we will include that as an axiomatic requirement.

2 Performance Metrics and their Properties

Given a trained model F_Θ and a new batch of data X, we wish to apply an operator $\bullet$ to revise configuration Θ, providing a new configuration, $\Theta \bullet X$ which provides a new model $F_{\Theta \bullet X}$ that has some desirable qualities that we will soon axiomitize in the next section. With that in mind, we will introduce a bit of the quantities we will analyze with respect to this framework:

1. Post-revision accuracy: $\mathbf{P}_{x \sim D}[F_{\Theta \bullet X}(x) = g(x)]$
2. Forgetting: $\mathbf{P}_{x \sim D}[F_{\Theta \bullet X}(x) \neq g(x) \mid F_\Theta(x) = g(x)]$

2.1 Q's for MVM from probability section

- In the probabilities in the axioms, should we make them where x is picked from X or D?
- Should we re-write the axioms to be about distribution D instead of batch X?

3 Revision Axioms

Definition 3.1 (Revision Axioms)**.** *Given model F_Θ with configuration $\Theta = \langle \Psi, \Pi \rangle$, batch X, operator $\bullet$, which returns revised configuration, $\Theta \bullet X = \langle \Psi', \Pi' \rangle$ and model $F_{\Theta \bullet X}$, we define the following revision axioms:*

1. *α-Success: Given $\alpha \in [0,1]$, $\mathbb{E}_{x \sim X}[F_{\Theta \bullet X}(x) = g(x)] \geq \alpha |X|$.*
2. *β-Inclusion: $\mathbf{P}_{x \sim X}[F_{\Theta \bullet X}(x) \neq g(x) \mid F_\Theta(x) = g(x)] < \beta$.*
3. *α-Vacuity: Givens $\alpha \in [0,1]$, if $\mathbf{P}_{x \sim X}[F_\Theta(x) = g(x)] \geq \alpha$ then for all $x \in \mathcal{X}$: $F_\Theta(x) = F_{\Theta \bullet X}(x)$.*
4. *Consistency: Π' is consistent.*

Proposition 3.1 (No Changes with Iterated Revisions)**.** *If $\bullet$ satisfies Axioms 1, 2, and 3 then $\Theta \bullet X$ bulletX $= \Theta \bullet X$.*

4 Revision Operator

Definition 4.1 (α-Fact Partition)**.** *For configuration Θ and batch X, we say partition (X_1, X_2) of X is an α-fact partition if:*

1. *there does not exist an $x_1 \in X_1$, $x_2 \in X_2$ s.t. $\mathcal{F}_\Theta(x_1) = \mathcal{F}_\Theta(x_2)$*
2. *and $\mathbf{P}_{x \sim X_1}[F_\Theta(x) = g(x)] \geq \alpha$.*

Definition 4.2 (Rule Learner)**.** *A rule learner rules is a function that maps a batch, ground truth function, and configuration to a set of rules in the language specified by atomic propositions specified by $\mathcal{P}$ and function symbols (treated as atomic propositions) where the bodies of the rules consist of conjunctions from $\mathcal{P}$ and the heads are function symbols specified by the model-query pairs in the configuration.*

Definition 4.3 (α-Splitting rule learner)**.** *Given a batch X, ground truth function g, and configuration Θ, rule learner rules is α-splitting w.r.t. X, g, Θ if there exists α-fact partition (X_1, X_2) of batch X such that $\{\Gamma_{\mathsf{rules}(X,g,\Theta)} \mathcal{F}_\Theta(x)\}_{x \in X_1} \cap \{\Gamma_{\mathsf{rules}(X,g,\Theta)} \mathcal{F}_\Theta(x)\}_{x \in X_2} = \emptyset$.*

Definition 4.4 (Strong α-Splitting rule learner)**.** *Given a batch X, ground truth function g, configuration $\Theta = \langle \Psi, \Pi \rangle$, model query function tuple $\langle f, q \rangle \notin \Psi$, and configuration $\Theta_{f,q} = \langle \Psi \cup \{\langle f, q \rangle\}, \Pi \rangle$, rule learner rules is strong α-splitting w.r.t. $X, g, \Theta_{f,q}$ if it is α-splitting w.r.t. $X, g, \Theta_{f,q}$ and $\{\Gamma_{\mathsf{rules}(X,g,\Theta_{f,q})} \mathcal{F}_\Theta(x)\}_{x \in X_2} = \{f\}$.*

Lemma 4.1. *Given a batch X, ground truth function g, configuration $\Theta = \langle \Psi, \Pi \rangle$, model query function tuple $\langle f, q \rangle \notin \Psi$, configuration $\Theta_{f,q} = \langle \Psi \cup \{\langle f, q \rangle\}, \Pi \rangle$, and strong α-splitting rule learner rules w.r.t. $X, g, \Theta_{f,q}$. If $\mathbf{P}_{x \sim X_2}[f(x) = g(x)] \geq \alpha$ then for operator $\bullet$ defined as $\Theta \bullet X = \langle \Psi \cup \{\langle f, q \rangle\}, \mathsf{rules}(X, g, \Theta_{f,q}) \rangle$ satisfies axiom 1.*

Proof. Need to write... □

Lemma 4.2. *Given a batch X, ground truth function g, configuration $\Theta = \langle \Psi, \Pi \rangle$, model query function tuple $\langle f, q \rangle \notin \Psi$, configuration $\Theta_{f,q} = \langle \Psi \cup \{\langle f, q \rangle\}, \Pi \rangle$, and strong α-splitting rule learner* rules *w.r.t. $X, g, \Theta_{f,q}$. If $\mathbf{P}_{x \sim X}[x \in X_1 \mid F_\Theta(x) = g(x)] \geq 1 - \beta$ then for operator $\bullet$ defined as $\Theta \bullet X = \langle \Psi \cup \{\langle f, q \rangle\}, \mathsf{rules}(X, g, \Theta_{f,q}) \rangle$ satisfies axiom 2.*

Proof. This requires us to show that $\mathbf{P}_{x \sim X}[f_{\Theta \bullet X}(x) \neq g(x) \mid F_\Theta(x) = g(x)] < \beta$, which is equivalent to $\mathbf{P}_{x \sim X}[f_{\Theta \bullet X}(x) = g(x) \mid F_\Theta(x) = g(x)] \geq 1 - \beta$. By the construction of $\bullet$, we can re-write the quantity $\mathbf{P}_{x \sim X}[f_{\Theta \bullet X}(x) = g(x) \mid F_\Theta(x) = g(x)]$ as the sum $\mathbf{P}_{x \sim X}[x \in X_1, F_\Theta(x) = g(x) \mid F_\Theta(x) = g(x)] + \mathbf{P}_{x \sim X}[x \in X_2, f(x) \neq g(x) \mid F_\Theta(x) = g(x)]$, which is equivalent to $\mathbf{P}_{x \sim X}[x \in X_1 \mid F_\Theta(x) = g(x)] + \mathbf{P}_{x \sim X}[x \in X_2, f(x) \neq g(x) \mid F_\Theta(x) = g(x)]$. As the statement of the Lemma establishes $\mathbf{P}_{x \sim X}[x \in X_1 \mid F_\Theta(x) = g(x)] \geq 1 - \beta$, this sum is also greater than or equal to $1 - \beta$, which completes the proof. □

5 Full Operator (rough sketch)

1. Given F_Θ and X, check the performance to determine if a change is required (Satisfying axiom 3)

2. We define $\Pi' = \mathsf{rules}_{pre}(X, g, \Theta)$ as a rule learner that returns the symbol error for certain samples such that the quantity $\mathbf{P}_{x \sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \Gamma_{\Pi'}(\mathcal{F}_\Theta(x)) \neq \mathsf{error}] - \mathbf{P}_{x \sim X}[F_{\Theta \bullet X}(x) = g(x)]$ is maximized subject to the constraint where $\mathbf{P}_{x \sim X}[F_{\Theta \bullet X}(x) \neq g(x) \ \textit{or} \ \Gamma_{\Pi'}(\mathcal{F}_\Theta(x)) = \mathsf{error} \mid F_\Theta(x) = g(x)] < \beta$

3. JUST DISCUSS KEEPIN Π small (old: rules_{pre} needs to be designed such that $|\Pi'| + |\Pi| \leq \epsilon |X|$ - this can probably be done with an additional cardinality constraint in the procedure of step 2.)

4. Define set $X_2 = \{x \in X \ s.t. \ \Gamma_{\Pi'}(\mathcal{F}_\Theta(x)) = \mathsf{error}\}$ and $X_1 = X \setminus X_2$.

5. $\langle f, q \rangle = \mathsf{tng}(X_2)$ - this should have α accuracy on X_2

6. Create Π'' based on the error rules in Π but replacing error in the head with f. The full $\mathsf{rules}(X, g, \Theta)$ returns $\Pi \cup \Pi''$. This should be provably consistent (satisfying axiom 4)

7. The operator returns configuration $\langle \Psi \cup \{\langle f, q \rangle\}, \Pi \cup \Pi'' \rangle$. Provided f achieves α accuracy and the maximized quantity from step 2 is at least α, we satisfy axiom 1 (Lemma 5.1); item 2 above and Lemma 5.2 lead to satisfying axiom 2; step 1 above satisfies axiom 3; step 6 above satisfies axiom 4; and the parsimony check satisfies axiom 5

6 Method

This section needs a lot of cleaning up...

6.1 Analysis of Post-Revision Accuracy

In what follows, we assume that there is a disjunctive set of formulas, frml that if $\mathcal{F}_\Theta(x) \models \mathsf{frml}$ that will cause the revision operator to rely on the non-revised model to evaluate sample x (i.e., $F_{\Theta \bullet X}(x) = F_\Theta(x)$ in this case). Otherwise, the revised model employs a different strategy to evaluate sample x (that is not F_Θ, i.e., we will use the symbol f^* as a place-holder). In short, define the revised model as follows:

$$F_{\Theta \bullet X}(x) = \begin{cases} f^*(x) & \text{if } \mathcal{F}_\Theta(x) \models \mathsf{frml} \\ F_\Theta(x) & \text{otherwise} \end{cases}$$

The idea here is the frml can serve as a metacognitive condition for deciding to keep the unrevised model.

Lemma 6.1. *Finding a disjunctive set of formulas frml that maximizes the quantity $Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \models \mathsf{frml}] - Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]$ is equivalent to finding the set M' that maximizes the quantity $Pr_{x\sim X}[F_{\Theta \bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \not\models \mathsf{frml}]$.*

Proof. First, we re-write the probability $Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \models \mathsf{frml}]$

$$\begin{aligned}
& Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \models \mathsf{frml}] && (1)\\
&= \frac{Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x), \mathcal{F}_\Theta(x) \models \mathsf{frml}]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (2)\\
&= \frac{Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} - \frac{Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x), \mathcal{F}_\Theta(x) \not\models \mathsf{frml}]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (3)\\
&= \frac{Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} - \frac{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}] && (4)\\
&= \frac{Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)] - Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}] Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (5)
\end{aligned}$$

This then allows us to re-write the quantity $Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \models \mathsf{frml}] - Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]$ as follows:

$$\begin{aligned}
& \frac{Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)] - Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}] Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}] - Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}] Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (6)\\
&= \frac{Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)] - Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}] Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}] - (1 - Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]) Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (7)\\
&= \frac{-Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}] Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}] + Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]) Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (8)\\
&= (-Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}] + Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)]) \frac{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (9)\\
&= (Pr_{x\sim X}[F_{\Theta \bullet X}(x) \neq g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}] + Pr_{x\sim X}[F_{\Theta \bullet X}(x) = g(x)] - 1) \frac{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} && (10)
\end{aligned}$$

Note that the quantity $Pr_{x\sim X}[F_{\Theta\bullet X}(x) = g(x)]-1$ can be treated as a constant, as it does not depend on frml, which allow us to re-write our objective quantity as:

$$Pr_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}]\frac{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]}{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathsf{frml}]} \tag{11}$$

$$= Pr_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}]\frac{Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]}{1 - Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}]} \tag{12}$$

Clearly, the resulting quantity can also be maximized by the following product:

$$Pr_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}]Pr_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}] \tag{13}$$

$$= Pr_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \not\models \mathsf{frml}] \tag{14}$$

Which give us the statement of the Lemma, completing the proof. □

Proposition 6.1. *The quantity $Pr_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \models \mathsf{frml}]$ is an increasing submodular function in frml.*

Proof. Need to prove....

□

6.2 Analysis of Forgetting

Lemma 6.2. $\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid F_\Theta(x) = g(x)] =$
$\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}, F_\Theta(x) = g(x)] \times \mathbf{P}_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml} \mid F_\Theta(x) = g(x)]$

Proof. Suppose, BWOC, the statement does not hold. Consider the produce $\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}, F_\Theta(x) = g(x)] \times \mathbf{P}_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml} \mid F_\Theta(x) = g(x)]$ which is equivalent to the following:

$$\frac{\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \not\models \mathsf{frml}, F_\Theta(x) = g(x)]}{\mathcal{F}_\Theta(x) \not\models \mathsf{frml}, F_\Theta(x) = g(x)]} \times \frac{\mathbf{P}_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml}, F_\Theta(x) = g(x)]}{\mathbf{P}_{x\sim X}[F_\Theta(x) = g(x)]} \tag{15}$$

$$= \frac{\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \not\models \mathsf{frml}, F_\Theta(x) = g(x)]}{\mathbf{P}_{x\sim X}[F_\Theta(x) = g(x)]} \tag{16}$$

$$= \mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \not\models \mathsf{frml} \mid F_\Theta(x) = g(x)] \tag{17}$$

However, by definition, we know that if $\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \models \mathsf{frml} \mid F_\Theta(x) = g(x)] = 0$. This means that $\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \not\models \mathsf{frml} \mid F_\Theta(x) = g(x)] = \mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid F_\Theta(x) = g(x)]$ which gives us a contradiction, proving the statement. □

Corollary 6.1. $\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid F_\Theta(x) = g(x)] \leq$
$\mathbf{P}_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml} \mid F_\Theta(x) = g(x)]$

Proof. This follows from Lemma 6.2 and the fact that $\mathbf{P}_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x) \mid \mathcal{F}_\Theta(x) \not\models \mathsf{frml}, F_\Theta(x) = g(x)] \leq 1$. □

Note that case where the bound is tight is when f^* (the new method of evaluating a sample employed by the revision) forgets all correct answers returned by F_Θ. In practice, we can set this to a constant (in our experiments we find $1-\alpha$ to be a good heuristic - with the intuition that the probability of forgetting is approximately the same as the probability of an incorrect answer, i.e. assuming that the errors of f^* are distributed the same across the correct and incorrect responses of F_Θ).

Proposition 6.2. *The quantity* $\mathbf{P}_{x\sim X}[\mathcal{F}_\Theta(x) \models \mathit{frml} \mid F_\Theta(x) = g(x)]$ *(equivalently* $1 - \mathbf{P}_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathit{frml} \mid F_\Theta(x) = g(x)]$ *is an increasing submodular function in* frml*.*

Proof. Need to prove □

6.3 Optimization Algorithm

As Proposition 6.1 showed that $1 - Pr_{x\sim X}[F_{\Theta\bullet X}(x) \neq g(x), \mathcal{F}_\Theta(x) \models \mathsf{frml}]$ is an increasing submodular function in frml and Proposition 6.2 $\mathbf{P}_{x\sim X}[\mathcal{F}_\Theta(x) \not\models \mathsf{frml} \mid F_\Theta(x) = g(x)]$ is also increaseing and submodular, we can frame this as a a special case of the "Submodular Cost Submodular Knapsack" (SCSK) problem and can be approximated with an efficient greedy algorithm [7].

Algorithm 1 DetRuleLearn

Require: $\alpha, \beta \in [0,1]$, Set of prompts $\mathcal{P}$
Ensure: Subset of prompts $\mathcal{P}'$
$\mathcal{P}' := \emptyset$
$\mathcal{P}'^* := \{c \in \ \ s.t.\ NEG_{\{c\}} \leq \epsilon \cdot \frac{N_i P_i}{R_i}\}$
while $\mathcal{P}'^* \neq \emptyset$ **do**
  $c_{best} = \arg\max_{c \in DC^*} POS_{DC_i \cup \{c\}}$
  Add c_{best} to DC_i
  $DC^* := \{c \in \backslash DC_i \ s.t.\ NEG_{DC_i \cup \{c\}} \leq \epsilon \cdot \frac{N_i P_i}{R_i}\}$
end while
return $\mathcal{P}'$

7 Misc. Notes

7.1 Some assumptions

- Training data is not accessible
- Batch X represents a new batch of data representative of data going forward. This is the "domain incremental" case where class labels stay the same and the distribution shifts gradually over time. The existing model will be good on some samples, but would be better off if re-trained.
- This version gives just one class per sample

7.2 Can the three probabilities be associated with the following quantities? Might be useful for proofs

- Accuracy: $\frac{|\{x \in X \ s.t. \ F_{\Theta \bullet X}(x) = g(x)\}|}{|X|}$
- Forgetting: $1 - \frac{|\{x \in X \ s.t. \ F_{\Theta}(x) = F_{\Theta \bullet X}(x) = g(x)\}|}{|\{x \in X \ s.t. \ F_{\Theta}(x) = g(x)\}|}$
- Revision FPR: $1 - \frac{|\{x \in X \ s.t. \ F_{\Theta}(x) \neq F_{\Theta \bullet X}(x) \ and \ F_{\Theta \bullet X}(x) = g(x)\}|}{|\{x \in X \ s.t. \ F_{\Theta}(x) \neq F_{\Theta \bullet X}(x)\}|}$

TODO: Check references for accuracy.... I used an LLM for the bibtex

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